COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-DBS-013: DBMS Server Inventory

Document ID	TS-DBS-013
Title	DBMS Server Inventory
Executive Sponsor	Database Services Management, DBS Manager
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT, state agencies as defined in C.R.S. 24-37.5-101(28), and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
Version	4.0
Effective Date	05-01-2021
Last Updated	07-01-2026

Purpose

This standard is issued under the authority of the Governor's Office of Information Technology (OIT)(C.R.S. 24-37.5, et seq.), Office of Enterprise Architecture (Rule 8 CCR 1501-7).

The purpose of this document is to communicate the standards that the Governor's Office of Information Technology (OIT) has developed with the intent of providing enterprise standard configurations for a Database Server Inventory. Standards are based on a best-of-value decision assessment involving multifarious factors, such as stakeholder needs, hardware configurations that are serviced and supported by OIT, as well as operational and security best practices guidelines.

The ultimate purpose of the Database Server Inventory standard is to maintain an accurate inventory of all the database instances and databases that the Database Services team supports.

Scope

These standards apply to OIT, state agencies as defined in C.R.S. 24-37.5-101(28) and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

The standards assigned and prescribed by this document, instituted for use by the approval of the OIT Architecture Review Board (ARB), should be used to govern purchase and deployment decisions deemed as “go-forward” or “net-new” strategies or implementations, and are not intended for use as retroactively enforceable mandate or policy for immediate conformance (especially in cases where cost implications may be prohibitive). This standard shall in no way require OIT to support any products and services not approved by OIT.

Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Update effective date and/or last review date.
- Publish the ARB approved Standard to the OIT public facing website.

Architecture Review Board (ARB)

- Engage stakeholders and subject matter experts to determine standard.
- Review/Approve Standard (this document).

Database Services Management - DBS Manager

- Owner of Standard.
- Engage stakeholders and subject matter experts to determine standard.
- Update document standards and version number.
- Present to the Architecture Review Board for approval and adoption.

DBS Team

- Baseline Documentation of all Database servers.
- Document changes to configuration after events that cause the changes, including but not limited to:
    - Installation of new databases or database servers
    - Changes in the environment affecting the DBMS or server.
    - Periodic maintenance such as patching
    - Ad-hoc changes – planned and unplanned.
- Make documentation available when needed.

Server Team

- Notify the DBMS team of potential or actual changes in the server environment so they can be documented.

Additional Assignments

All OIT staff are responsible to be aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

Standards

- Each database system configuration will be inventoried.
- Baseline Configuration: A baseline configuration inventory of all Database Servers will be created and stored.
    - All baseline configuration inventories that can be discovered in ServiceHub will be maintained in the ServiceHub CMDB tables. Baseline configuration inventories of database platforms not discoverable in ServiceHub will be stored in a secure, central repository.

- Inventory data discovered by ServiceHub will be verified and documented in a consistent way across agencies.
    - Only DBA team members and other authorized OIT personnel will have access to the configurations.
    - DBA staff will identify what elements will be included in the inventory and how these elements will be collected.
    - The elements must meet the requirements of all applicable OIT/State/Federal policies for configuration and asset management.
- The inventory must be kept as updated as possible by identifying specific activities that result in changes to the elements in the inventory. Some of these activities include but are not restricted to:
    - Lifecycle: installing a new database server instance, creating/installing a new database, routine patching and updates, upgrading to new database server versions, decommissioning a database, instance, or server
    - Application changes: design changes, new features, application migration to new technology
    - Performance management, monitoring, and troubleshooting: bug fixes, disaster recovery, routine monitoring reveals issues
- Wherever possible automated processes and tools should be used to collect and update inventories and that these tools be kept in a common repository and only be available to DBAs and other authorized OIT personnel.
- Tools:
    - ServiceHub CMDB discovery
    - Scripts (e.g. PowerShell, Python)
    - Database platform-specific scripts (e.g. TSQL for SQL Server, PL/SQL for Oracle)
    - 3rd party applications

Exceptions and Variance

Requests for variances from any provisions with this Standard must be requested using the Technical Standard End User Exception Request Procedure.

The purpose of the Exception Request process is to address business or technical reasons for the inability to adhere to one or more requirements in this standard.

Requests for exceptions should include compensating controls, that is, processes, technologies, or procedures proposed used to mitigate any deviation from the standard requirement. The Secure Configuration Exception Request form must be submitted to oit_ciso@state.co.us. Exception requests will be processed and recommended for approval, conditional approval, or denial.

References

OIT documents and policies

- Colorado Information Security Policies (CISPs)
- Enforcement Category Chart
- TS-CISO-001 – Data Security Categorization

Definitions

Standard: A rule that specifies a particular course of action or response to a given situation. Standards are mandatory directives to carry out management's policies and are used to measure compliance with policies. Standards serve as specifications for the implementation of policies and are designed to promote implementation of high-level organization policy rather than to create new policy in themselves.

CMDB: ServiceHub's Configuration Management Database (CMDB) is a centralized repository that acts as a data warehouse for information technology (IT) installations. It is designed to store and manage data related to IT assets, including hardware, software, networks, facilities, and personnel, as well as the relationships between these assets. The CMDB allows organizations to maintain a detailed inventory of their IT environment and provides a comprehensive view of how these components interact.

DBS: Database Services.

Database Server: An interconnected system of hardware and DBMS software designed to manage collection of data in such a manner as to maintain the confidentiality, integrity and availability of said data.

Host Server: The hardware/software that hosts a Database server.

Database: A relational collection of tables, stored procedures, views, indexes, constraints, jobs, maintenance plans that contain and manage a discrete set of data, i.e. for a single application.

Data Classification: The process of defining a collection of data based on an assessment of the potential impact on State organizations, assets or operations that a potential loss of confidentiality, integrity, or availability might have. See Technical Standard TS-CISO-001.

Element: A single piece of information about the configuration of a database system that is collected during the inventory process.

Element Category: The type element information such as database server, database, Host server.

OEA (Office of Enterprise Architecture): The Office of Enterprise Architecture was formed as an office under OIT by Governor's Rule 8 CCR 1501-7.

OIT (Office of Information Technology): The Governor's Office of Information Technology was created and governed by C.R.S. 24-37.5, et seq.

Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification for granting the exception. Typically an exception request is subject to review and approval by an authority in the area.